FUSED INPUT, PROTECTION AND 5 V SUPPLY

C1
VIN_FB
GND
10u 63V EEEFK1J10GP

C2
VIN_FB
GND
2.2u 100V X7R

C3
VIN_FB
GND
2.2u 100V X7R

U1
GND
VIN_FB
VOUT
RON
FB
PGOOD
BST
SW
GND
LM5164QDDANQ1

L1
SW
VS_LOGIC
68u WSS1346T-68SHLC

R1
RON
GND
41.2k

R2
VIN_FB
VALID
1M

R3
VALID
GND
221k

R4
VS_LOGIC
FB
156k

R5
FB
GND
49.9k

R6
SW
RIPPLE
150k

C4
BST
SW
2.2n 50V X7R

C5
RIPPLE
VS_LOGIC
3.3n 50V X7R

C6
RIPPLE
FB
180p C0G

C7
VS_LOGIC
GND
22u 25V X7R

C8
VS_LOGIC
GND
22u 25V X7R

R7
VS_LOGIC
PGOOD
100k

F1
VIN_ACC
ACC_FUSED
3A MINI MAIN / LED

F2
LED_POS
VIN_PROT
1A MINI LOGIC

U4
DGATE
ACC_FB
ACC_SW
ON_OFF
ON_OFF
POWER_MID
VIN
HGATE
LED_POS
POWER_MID
CAP_PUMP
POWER_MID
GND
LM74480-Q1

D1
ACC_FB
GND
TPSMB33CA

Q5
ACC_FB
ACC_FB
ACC_FB
HGATE
POWER_MID
BUK9Y15-100E

Q6
LED_POS
LED_POS
LED_POS
HGATE
POWER_MID
BUK9Y15-100E

R21
ON_OFF
ON_SENSE
46.44 0.1%

R22
ON_SENSE
GND
4.02k 0.1%

R23
HGATE
VANUSH
100R

C13
ACC_FB
GND
100n 100V X7R

C14
POWER_MID
GND
100n 100V X7R

C15
CAP_PUMP
POWER_MID
100n 50V X7R

C16
VANUSH
GND
10n 100V X7R

C17
LED_POS
GND
47u 50V

PWR_FLAG
GND

PWR_FLAG
VIN_ACC

PWR_FLAG
ACC_FUSED

PWR_FLAG
LED_POS

PWR_FLAG
VIN_PROT

PWR_FLAG
VS_LOGIC

VEHICLE INPUT AND CONTROLLER STACK

H1
PWM_B
PWM_A
VCL_NOMINAL
PWM_B

H2
ENABLE_REQ
ICH_NOMINAL
GND

H3
ESP_5V
GND
GND
POWER_NOMINAL

H4
PWM_B
AD_NOMINAL

TP5
VS_LOGIC
VS_LOGIC

TP6
GND
GND

TP7
PGOOD
PGOOD

TP8
ENABLE_REQ
ENABLE_REQ

J1
GND
VIN_ACC
VEHICLE IN

J2
LED_W_NEG
RGBW OUT

JP1
VS_LOGIC
ESP_5V
5V RUN SHUNT

MH1
M3 CASE SUPPORT

MH2
M3 CASE SUPPORT

MH3
M3 CASE SUPPORT

MH4
M3 CASE SUPPORT

ONE COMMON-POSITIVE RGBW ZONE

U2
ENABLE_REQ
PWM_B
GND
PWM_B
DRV_G
VS_LOGIC
DRV_B
ENABLE_REQ
UCC27524ADDRQ1

U3
ENABLE_REQ
PWM_B
GND
PWM_B
DRV_W
VS_LOGIC
DRV_R
ENABLE_REQ
UCC27524ADDRQ1

C9
VS_LOGIC
GND
100n 25V X7R

C11
VS_LOGIC
GND
100n 25V X7R

C10
VS_LOGIC
GND
1u 25V X7R

C12
VS_LOGIC
GND
1u 25V X7R

R8
ENABLE_REQ
GND
2.2k

Q1
GND
GND
GND
GATE_R
LED_R_NEG
BUK9Y15-60E

R9
DRV_R
GATE_R
100

R13
GATE_R
GND
10k

R17
PWM_R
GND
100k

D3
LED_RGBW
GND
SMBJ24CA

Q2
GND
GND
GND
GATE_G
LED_G_NEG
BUK9Y15-60E

R10
DRV_G
GATE_G
100

R14
GATE_G
GND
10k

R18
PWM_G
GND
100k

D4
LED_RGBW
GND
SMBJ24CA

Q3
GND
GND
GND
GATE_B
LED_B_NEG
BUK9Y15-60E

R11
DRV_B
GATE_B
100

R15
GATE_B
GND
10k

R19
PWM_B
GND
100k

D5
LED_RGBW
GND
SMBJ24CA

Q4
GND
GND
GND
GATE_W
LED_W_NEG
BUK9Y15-60E

R12
DRV_W
GATE_W
100

R16
GATE_W
GND
10k

R20
PWM_W
GND
100k

D6
LED_W_POS
GND
SMBJ24CA

EVALUATION HARDWARE / VERIFY VEHICLE FEED, LED PINOUT AND USB ISOLATION

Sheet: /
File: vhos_rgbw_revb.kicad_sch
Title: 4Runner two-connector RGBW shield
Size: A1 | Date:
KiCad E.D.A. 10.0.6
Rev: B
Id: 1/1